IAN SNIDER

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Nuclear physicist skilled in autonomous control systems, HPC, machine learning, and nuclear data, seeking to address national security concerns by improving our fundamental understanding of neutron reactions.

EDUCATION

Graduate Researcher in Nuclear Engineering - University of California, Berkeley *Aug 2025 - present*

B.S. Mechanical Engineering - Washington University in St. Louis 4.00/4.00 *May 2025*

B.A. Physics - Truman State University 3.89/4.00 *Dec 2024*

PROJECTS

WashU Robotics MATE ROV & MARINER Projects - ***Mechanical Lead*** *2023 - 2025*

- Collaborated with other students to develop an advanced autonomous underwater vehicle (AUV)
- Prototyped hydrodynamic dive control using an IMU and sonar state estimation with a Kalman Filter
- Built a chassis and buoyancy engine

ESE 4481 Autonomous Aerial Vehicle Control - ***Student*** *Spring 2025*

- Designed and succesfully implemented a cascaded PID/PD control system on a CrazyFlie 2.1 quadcopter

EXPERIENCE

Lawrence Berkeley National Laboratory - *Lab Affiliate*, *Berkeley, CA* *Aug 2025 - present*

- Graduate researcher in the Bay Area Neutron Group studying experimental techniques for indirect cross-section measurement of short-lived nuclei

Lawrence Livermore National Laboratory - *Graduate Intern*, *Livermore, CA* *June 2025 - present*

- Developed CRISP (CRItical Simulation Pipeline) for parallelizing jobs from the radiation transport code: COG
- Test driven Python development included Poetry, CI/CD, SQLite, MPI, CLI, feature branching, and Sphinx
- Used 2 world-class LLNL HPC clusters Dane and Ruby with Slurm sbatch/srun scripts
- CRISP sucessfully automated the materials compiling and k_{eff} benchmarking of ~3420 ICSBEP critical reactors
- Maintainer of the CRISP project

Brookhaven National Laboratory - *Nuclear Science Intern*, *Upton, NY* *Summers 2022 - 2024*

- Developed BRR (Bayesian Resonance Reclassifier) for resonance reclassification on National Nuclear Data Center clusters
- Applied machine learning using Scikit-learn and Pytorch to classify resonance spin-group assignments for capture cross-sections of In-115, Pb-206, and U-238
- Used Random Matrix Theory and nuclear data to create synthetic training sets
- Used the neutron transport code OpenMC to perform perturbative sensitivity analyses of the thermal $1/v$ neutron capture cross-sections for critical reactors on Gd-155, 157 and U-235, 238

Truman State University - *Student Astronomy Researcher*, *Kirksville, MO* *2021 - 2022*

- Calculated trajectories for ~3000 Starlink satellites to optimize telescope viewing plans and developed a GUI

SKILLS

- **Programming:** C, C++, Python, Go, Bash, Lua, MATLAB, Mathematica, Octave, LaTeX, Typst, Vimscript
- **Software/Technical:** Simulink, SolidWorks, OpenMC, COG, MCNP, Pytorch, Scikit-learn, Slurm, Git, Linux, Arduino
- **Physics/engineering:** Autonomous Aerial Vehicle Control, Classical Mechanics, Electrodynamics, Electronics, Vibrations, Quantum Mechanics, Mathematical Physics, Nuclear Physics, Fluid Mechanics, Solid Mechanics, Acoustics, Materials Science, Thermal Systems, Criticality Safety, Radiation Biology, Turbojets, Ramjets
- **Math:** Linear Algebra, ODEs, Computing Structures, Control Systems, Machine Learning, Monte Carlo, and Optimizations

ACTIVITIES

WashU Climbing - ***Member*** *2023 - 2025*

- Indoor and outdoor bouldering and leading climbing

Society of Physics Students - ***Demo Chair*** *2020 - 2023*

- Developed 3 new physics demos, performed demos at meetings, and provided volunteer physics tutoring
- Wrote and proctored exams for 2022 & 2023 Science Olympiads ("Crave the Wave" and "Remote Sensing")

TECHNICAL REPORTS AND ABSTRACTS

[1] G. Nobre *et al.*, "Annual Report on NCSP Technical Support task in BNL during FY24," 2024. doi: 10.2172/2474826.

[2] I. Snider, G. Nobre, and D. Brown, "Resonance Capture Widths for the Bayesian Resonance Reclassifier," in *APS Meeting Abstracts*, in APS Meeting Abstracts. Jan. 2023, p. DB3.074. [Online]. Available: https://ui.adsabs.harvard.edu/abs/2023APS..HAWD03074S

[3] I. Snider, G. Nobre, D. Brown, and W. Fritsch, "Accuracy Correlation in Neutron Resonance Reclassification," *Bulletin of the American Physical Society*, vol. 67, 2022.

CONFERENCE PRESENTATIONS

- Lawrence Livermore National Laboratory Student Research Conference. Lawrence Livermore National Lab B511, Livermore, CA, August 7th, 2025.
- Brookhaven National Laboratory Student Research Conference. Brookhaven National Lab Bldg. 488, Upton, NY, August 9th, 2024.
- American Physical Society - Division of Nuclear Physics and Japan Physical Society joint fall meeting. Hilton Waikoloa Village, The Big Island, HI, Nov 27-Dec 1, 2023.
- Brookhaven National Laboratory Student Research Conference. Brookhaven National Lab Bldg. 488, Upton, NY, August 10th, 2023.
- Truman State University Student Research Conference. Truman State University, Kirksville, MO, April 21st, 2023.
- American Physical Society - Division of Nuclear Physics fall meeting. Hyatt Regency Hotel, New Orleans, LA, October 29-31, 2022.
- Brookhaven National Laboratory Student Research Conference. Brookhaven National Lab Bldg. 488, Upton, NY, August 11th, 2022.

HONORS

Kenneth L. Jerina Prize for Outstanding Dual-Degree in Mechanical Engineering - WashU Engineering *April 2025*

- Honored for academic achievements and engineering community contributions.

Conference Experience for Undergraduates 2023 - APS DNP *September 2023*

- Competitive research abstract award
- Invitation to present a research poster at the APS DNP Fall 2023 conference on The Big Island, HI

Conference Experience for Undergraduates 2022 - APS DNP *August 2022*

- Competitive research abstract award
- Invitation to the poster presentation at the APS DNP Fall 2022 meeting in New Orleans, LA

Sigma Pi Sigma Honor Society - Truman State University *May 2022*

- Recognized for service and academic scholarship in physics